

Randomized and low-rank approximation

Sharp joint rank and accuracy dependence for strong ℓ_p subspace coresets

Difficulty: challenging **Importance:** interesting to the community**Status:** Solved

Rating rationale: Challenging because simultaneous control of all subspaces must match joint rank and accuracy lower bounds; community impact is compact robust low-rank fitting.

Full resolution — even/non-even classification, 14 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Verified affiliation and submission record.](#)

[Part I, Theorem 1.1](#), and [Part II, Theorem 1.1](#) classify the optimal worst-case strong original-row coreset size for every fixed real $p > 2$, all $k \geq 1$ and $0 < \varepsilon < 1/2$, with arbitrary input rank and ambient dimension. Writing $S_p(k, \varepsilon)$ for that size, the result is

$$S_p(k, \varepsilon) = \tilde{\Theta}_p \left(\begin{cases} k^{p/2} \varepsilon^{-2}, & p \notin 2\mathbb{Z}, \\ \min\{k^{p/2} \varepsilon^{-2}, k^{(p+1)/2} \varepsilon^{-1} + k^{p/2-1} \varepsilon^{-2}\}, & p \in 2\mathbb{Z}. \end{cases} \right)$$

The hidden factors are logarithmic in $2k/\varepsilon$; constants may depend only on the fixed exponent. Weights are nonnegative and supported on original rows, and the guarantee holds simultaneously for every fitted subspace of dimension at most k . The displayed additive proposal in the retained problem is false for every fixed $p > 2$. This completes the broader classification as well as its subsidiary truth question. The general upper bound for non-even powers is credited to Lin–Mirrokni–Woodruff; the manuscript credits Li–Wang–Woodruff for the antecedent Fourier mechanism.

Both complete proof parts passed a separate [independent Codex AI-agent informal audit](#), including the new uniform estimates and primary-source dependency checks. This meets the repository’s **Solved** policy. The exact cubic certificate passed 17,575 assertions; optional numerical diagnostics were not rerun because their dependencies were unavailable. The proof audit, rather than those finite checks, supports the universal result. [Part I source](#) · [Part II source](#).

ChatGPT assistance is disclosed. No Lean verification, external human peer review, formal verification or priority certification is asserted. The earlier partial results and historical open-status checks below are retained as history; this dated full resolution supersedes their remaining-gap statements. Original ID, canonical path and mathematical target are unchanged.

Further partial resolution — even-power high accuracy, 13 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Submission and verified affiliation.](#)

For each fixed integer $s \geq 2$, [Theorems 1.1–1.2](#) and [Corollary 1.3](#) establish

$$c_s \frac{k^{s-1}}{\varepsilon^2} \leq S_{2s}(k, \varepsilon) \leq C_s \left(\frac{k^{(3s-1)/2}}{\varepsilon} + \frac{k^{s-1}}{\varepsilon^2} \right) \log^5 \left(\frac{2k}{\varepsilon} \right).$$

Thus $S_{2s}(k, \varepsilon) = \tilde{\Theta}_s(k^{s-1}/\varepsilon^2)$ whenever $0 < \varepsilon < 1/2$ and $\varepsilon \leq k^{-(s+1)/2}$. The bounds hold for every $k \geq 1$, with arbitrary input rank and ambient dimension, nonnegative original-row weights, and simultaneous preservation of every subspace of dimension at most k . Constants depend only on s ; no efficient construction is claimed.

The stated partial results passed an [independent Codex AI-agent informal audit](#), including primary-source dependency checks and 8,453 finite assertions. Finite checks are supporting diagnostics, not proofs of the asymptotic theorem. ChatGPT assistance is disclosed; no external human peer review or formal verification is asserted. No Lean verification was performed.

RA-05 remains Partially resolved. Non-even exponents and intermediate accuracies for even $p \geq 6$ remain unclassified. The earlier quartic classification and all-exponent lower bounds below are retained as separate prior contributions. [New proof source](#).

Partial resolution — unrestricted quartic case, 13 September 2026

Author: Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation. [Submission and verified affiliation](#).

Theorem 1.1, proved in Sections 2–8 classifies the optimal size for $p = 4$, arbitrary input rank, every integer $k \geq 1$ and every $0 < \varepsilon < 1/2$, up to logarithmic factors:

$$S_4(k, \varepsilon) = \tilde{\Theta}\left(\min\left\{\frac{k^2}{\varepsilon^2}, \frac{k^{5/2}}{\varepsilon} + \frac{k}{\varepsilon^2}\right\}\right).$$

The weights are nonnegative and select original rows; preservation is simultaneous over all subspaces of dimension at most k , without a restriction on input rank, row count or ambient dimension. The lower bound has no logarithmic loss, the combined upper bound uses at most $\log^9(2k/\varepsilon)$, and the second upper branch uses at most the fifth power. [Proof source](#).

This also disproves the subsidiary displayed upper-bound conjecture below at $p = 4$: at $\varepsilon = k^{-1}$ the lower bound is of order $k^{7/2}$, exceeding its proposed order k^3 times any fixed logarithmic power. **The unrestricted optimal joint classification for fixed real $p > 2$ other than 4 remains open.** The whole RA-05 entry is therefore Partially resolved, and remains in the open count.

The argument passed a separate [independent Codex AI-agent informal audit](#), including primary-source checks of the imported theorems. All 7,621 finite assertions and both exact certificate checkers were rerun successfully. ChatGPT assistance is disclosed; this is not external human peer review or formal verification. No Lean verification was performed. The [earlier all-exponent lower-bound submission, PR #199](#), is a separate partial contribution. The original statement, ID, path and prior-source credit are retained below.

All-exponent lower bound — 13 September 2026

Holden's separate [Theorem 1.1](#) proves, for every fixed real $p > 2$, all sufficiently large k and every $0 < \varepsilon < 1/2$,

$$S_p(k, \varepsilon) \geq c_p \frac{k^{p/2}}{\varepsilon^{\beta_p} + (\log k)/k}, \quad \beta_p = \begin{cases} 2, & p \in \{4, 6, 8, \dots\}, \\ 2 - 2/p, & \text{otherwise.} \end{cases}$$

One input per rank, independent of accuracy, witnesses the bound for arbitrary nonnegative original-row weights. Corollary 1.2 disproves the subsidiary additive formula for **every fixed real exponent greater than two**, even allowing any fixed logarithmic power. Corollary 1.3 matches the cited upper bound up to logarithms for even exponents at least four when $\varepsilon \geq \sqrt{(\log k)/k}$. The [independent review and submission record](#) are retained. Together with the quartic classification above, these results leave the other exponents' unrestricted optimal joint sizes unresolved.

Original question

Fix a real $p > 2$. For $A \in \mathbb{R}^{n \times d}$, with rows a_i^T , and a linear subspace $F \subseteq \mathbb{R}^d$, define

$$\text{cost}_A(F) = \sum_{i=1}^n \|a_i^T(I - P_F)\|_2^p,$$

where P_F is the Euclidean orthogonal projector. A strong row coreset consists of nonnegative weights $w_1, \dots, w_n$, supported on a small subset of the original rows, for which

$$(1 - \varepsilon) \text{cost}_A(F) \leq \sum_i w_i \|a_i^T (I - P_F)\|_2^p \leq (1 + \varepsilon) \text{cost}_A(F)$$

holds simultaneously for every F of dimension at most k .

Determine the optimal worst-case coreset size as a joint function of k and ε , up to logarithmic factors. In particular, do constants $C_p, c_p > 0$ exist such that, for all n, d , integers $1 \leq k < d$, all input matrices A , and all $0 < \varepsilon < 1/2$, such a coreset exists with

$$|\text{supp}(w)| \leq C_p \left(\frac{k^{p/2}}{\varepsilon} + \frac{k}{\varepsilon^2} \right) \log^{c_p} \left(\frac{2k}{\varepsilon} \right) ?$$

The constants may depend on fixed p , but not on n, d, k, ε . This is an existence/size question; no extra running-time target is imposed.

Such a coreset permits subsequent robust low-rank fitting on a much smaller weighted matrix. The August 2026 paper proves an upper bound of order $\widetilde{O}_p(k^{p/2}/\varepsilon^2)$ and explicitly identifies the remaining gap to the joint lower-bound expression in Section 1.4. Its new result supersedes older questions that merely asked for an ε^{-2} dependence.

References

1. H. Lin, V. Mirrokni, and D. P. Woodruff, *Nearly Optimal Strong Coresets for ℓ_p Subspace Approximation*, arXiv:2608.26047v2, Section 1.4 “Open problems”; Section 1.2 for the new bounds. [Paper](#).
2. D. P. Woodruff and T. Yasuda, *Root ridge leverage score sampling for ℓ_p subspace approximation*, FOCS 2025, full version arXiv:2407.03262. [Paper](#).

Status check — 2026-09-08

Searched “strong coresets subspace approximation p greater than 2 epsilon 2026” and the exact 2026 title; checked the latest arXiv v2 of August 27, 2026. No later resolution was found. The target comes from the remaining question in that paper, rather than its already improved predecessor bounds.

Audit — 2026-09-10

Rechecked [Lin–Mirrokni–Woodruff, §1.4](#) and the August 27 version record. The joint k, ε gap remains explicit. Strong-coreset follow-up searches found no resolution. Difficulty was raised because the target concerns all input matrices and requires a new uniform sampling analysis.